

2.5 Detailed specific example: The Great Barrier Reef, Australia



The Great Barrier Reef is the world's largest coral reef system, stretching over 2300 km along the northeast coast of Australia. It is composed of over 2900 individual reefs and 900 islands, covering an area of 350 000 square km.

The importance of the Great Barrier Reef

The Great Barrier Reef is the largest living structure on Earth and is a significant economic, social, cultural, and environmental asset for Australia. It contributes over six billion dollars annually to Australia's economy through industries such as tourism and fishing. For example, over two million people visit the reef each year, taking part in activities like sailing, snorkelling, and scuba diving. The reef supports various economic industries and provides employment for local people. Over 64 000 people earn a living related to the reef. This is more than the total number of employees of Australia's major banks.

It also holds cultural significance for the indigenous peoples of Australia, particularly the First Nations peoples. Much of their culture and traditional livelihoods have been built around its rich ecosystem over the last 60 000 years.

The Great Barrier Reef is **biodiverse**, with over 1500 species of fish and 450 species of hard coral. Biodiversity is important to the reef because each species contributes to the overall stability of the reef. For example, predators like sharks keep the numbers of coral-eating fish under control. Despite occupying less than 1% of the ocean floor, coral reefs are home to more than 25% of all marine life on Earth. The value of the reef was confirmed when, in 1981, the Great Barrier Reef was selected as a World Heritage Site. This means that the reef is considered to be of outstanding value to humanity and is legally protected.



Figure 2.18: The Great Barrier Reef in Australia is the largest coral reef system in the world



Figure 2.19: A scuba diver explores the Great Barrier Reef

TASK 1

- a** Using the data in Table 2.5, **draw a bar graph** to show the number of tourists visiting the Great Barrier Reef over time.
- b** **Describe the graph.**
- c** How do you think the effects of climate change will impact the number of tourists visiting the Great Barrier Reef? Suggest how this can be managed in the short-term and long-term.

Table 2.5: Number of tourists visiting the Great Barrier Reef over time

Financial year	Number of tourists visiting the Great Barrier Reef (millions)
2013 to 2014	1.3
2014 to 2015	1.4
2015 to 2016	1.6
2016 to 2017	1.6
2017 to 2018	1.7
2018 to 2019	1.6
2019 to 2020	1.1
2020 to 2021	0.6
2021 to 2022	0.8
2022 to 2023	1.3

Threats to the Great Barrier Reef

The Great Barrier Reef faces numerous threats, both physical and human.

The crown-of-thorns starfish (COTS) poses a significant threat to the Great Barrier Reef. These starfish eat the living coral. This leaves the reef white in colour, similar to coral bleaching. A COTS outbreak is classified as 15 or more starfish per hectare of the reef. Each starfish can eat up to ten square metres of coral a year. COTS outbreaks can be made worse by human activity. Pollution from land-based runoff can cause algal blooms that provide a boost in food source for COTS.

Coral bleaching occurs when corals expel the symbiotic algae living in their tissues, which gives the corals their vibrant colours. This is often in response to environmental stresses such as increased sea temperatures due to, for example, a heatwave. The loss of these algae leads to a whitening, or bleaching, of the coral. Depending on the intensity of a heatwave, coral bleaching can occur quickly over a number of weeks, or slowly over several months. The Great Barrier

Reef has experienced several significant bleaching events in recent decades, with notable events outlined in Table 2.6. More recent events were particularly severe, affecting large portions of the reef. While bleached corals are not dead, they are significantly weaker and more susceptible to disease.



Figure 2.20: A crown-of-thorns starfish feeds on stony coral on the Great Barrier Reef



Figure 2.21: Coral bleaching on the Great Barrier Reef

Pollution from the land, such as agricultural runoff, adds excess nutrients like nitrogen and phosphorus. These pollutants smother the coral and lead to algal overgrowth which disrupts coral growth. Pollution from ships, such as oily ballast water, can also smother coral and spread diseases.

The impacts from overfishing on coral reef areas can lead to the depletion of key reef species. This can have an economic effect on local people who rely on these fish as a source of income. Related activities like anchoring and dredging can also damage the coral.

TASK 2

In pairs, evaluate the significance of the threats to the Great Barrier Reef. Which do you think has the greatest potential to damage the reef? Which has the least? Justify your answers.

The Great Barrier Reef Marine Park Authority was established in 1975. It is an Australian environmental government agency that is responsible for the management and protection of the Great Barrier Reef Marine Park. The authority’s role includes developing and implementing sustainable management plans, issuing permits for activities within the marine park, conducting scientific research, and monitoring the health of the reef.

Many of the threats to the Great Barrier Reef are global ecological issues, like climate change. As such, many of the strategies to protect the reef are connected to efforts to address climate change. These include international agreements to limit greenhouse gas emissions as well as local initiatives such as a transition to renewable energy sources.

Techniques and sustainable management

In order to manage the reef, one technique has been to divide the park into different zones. Each zone has different levels of protection and activity restrictions. This aims to balance conservation goals with sustainable use. For example, some areas are designated as no-entry zones to protect sensitive habitats, while others allow for activities like fishing and tourism under certain conditions.



Source A: An example of zoning on the Great Barrier Reef identifying the activities that are, or are not, permitted in each area

Due to its global significance, managing the Great Barrier Reef involves collaboration between various stakeholders. This includes government agencies, scientific institutions, indigenous communities, the tourism industry, and environmental organisations. Examples of this include the collaborative efforts between government agencies such as the Australian Department of the Environment and Energy, which establishes policies and regulations, and scientific institutions like the Australian Institute of Marine Science, which contributes valuable research and expertise. Researchers are investigating the genes of heat-tolerant corals which will provide an understanding as to how and where corals might be able to resist warming temperatures. The tourism industry actively participates in reef preservation by adhering to sustainable practices such as litter picks, education programmes, and managing COTS numbers. Environmental organisations like the Great Barrier Reef Foundation play a crucial role in fundraising and implementing conservation initiatives such as coral planting.

TASK 3: Study Source A

a

Identify potential conflicts between stakeholders and other users of the reef. Explain the causes of these potential conflicts.

b

Suggest what the Great Barrier Reef Marine Park Authority could do to alleviate these conflicts.

In addition to preventing further harm, sustainable management can include techniques for addressing existing damage. Recovery from coral bleaching events depends on sea temperature reduction or, in some cases, adaptation of the coral. Scientists and conservationists are conducting coral restoration programmes that involve growing new coral in controlled conditions and transplanting it back on to the reef. The Great Barrier Reef Foundation identifies areas of the reef that are damaged and plants new corals to speed up the recovery process. Researchers are also attempting to breed heat-tolerant corals to ensure the reef can cope with rising sea temperatures. However, these are both short-term solutions whilst the world looks to reduce its carbon emissions. The sustainable management of coral reefs requires both short-term and long-term solutions.

Table 2.6: A timeline of bleaching events and management steps for the Great Barrier Reef

1975	The Great Barrier Reef Marine Park Authority is established by the Australian government to preserve and protect the Great Barrier
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	Reef.
1981	The Great Barrier Reef is listed as a World Heritage Site. This means the reef is given legal protection due to its outstanding value to humanity.
1998	Coral bleaching event that affected 74% of inshore reefs and 21% of offshore reefs on the Great Barrier Reef.
1999	The non-profit Great Barrier Reef Foundation is established to help protect and preserve the reef.
2002	Coral bleaching event that saw 54% of corals bleached to some extent, with 18% strongly affected on the Great Barrier Reef.
July 2015	A UNESCO (a UN agency) committee decides against putting the Great Barrier Reef on the ‘in danger’ list, but stresses that Australia must do more to protect the reef.
2016	Mass coral bleaching affecting the northern end of the Great Barrier Reef. 30% of the reef’s total coral dies.
2017	Severe coral bleaching affecting the central third of the Great Barrier Reef.
April 2018	The Australian government gives \$443 million to the Great Barrier Reef Foundation.
August 2019	The Great Barrier Reef Marine Park assessment finds that the reef’s five-year outlook has moved from ‘poor’ to ‘very poor’.
March 2020	Third mass bleaching event in five years. 25% of reefs surveyed had ‘severe’ (more than 60% of coral bleached) levels of coral bleaching.
February 2021	Australian government report finds the marine environment along the Great Barrier Reef to be in ‘poor health’.
June 2021	UNESCO’s world heritage centre recommends the Great Barrier Reef is listed as a world heritage ‘in danger’ due to Australia’s reluctance to reduce its greenhouse gas emissions.
March 2022	Aerial surveys confirm the fourth mass bleaching event to impact the Great Barrier Reef in seven years. ‘Severe’ bleaching affects 43% of the reefs surveyed. The worst affected is the central region of the reef.
August 2023	UNESCO recommends the Great Barrier Reef not be placed on the list of world heritage sites that are ‘in danger’ but recognises that the reef remains under serious threat.

TASK 4

Study the timeline in Table 2.6. Evaluate which event you think has had the most significant impact in managing the Great Barrier Reef. Is this a positive or negative impact?

REFLECTION

When revising a detailed specific example, what strategies help you remember important content? What strategies do you find least helpful? Share your ideas with a partner.

TASK 5

- a Research online what it means for a World Heritage Site to be listed as ‘in danger’.
- b Study Source B. In pairs, discuss the potential problems of not listing the Great Barrier Reef as ‘in danger’.

No UN ‘In Danger’ Listing for Australia’s Great Barrier Reef

Marine experts are warning that Australia needs to do more to protect one of its greatest natural treasures after the United Nations recommended the Great Barrier Reef not be placed on a list of world heritage sites “in danger.” The world’s largest coral system is facing a range of threats, including climate change, pollution, and overfishing.

United Nations scientific advisors said that Australia has taken positive steps to protect the Great Barrier Reef since a UN monitoring mission visited Queensland in March 2022. It won’t—for now—be listed as a World Heritage site “in danger”.

Source B: A news article about the Great Barrier Reef from VOA News

IMPROVE THIS ANSWER

For a named area you have studied, explain how coral reefs can be managed sustainably.

In Australia 1 they 2 have managed the reef by splitting it into zones that only allow certain activities to take place. 3 They have collaborated with local people. The reef was listed as a World Heritage Site in 1981.

In places, tourists are asked not to stand on the coral. In 2019 the reef suffered a mass bleaching event. 4 The reef has also suffered from COTS 5 which eat the coral. In April 2018, the Australian government gave 443 million dollars to the Great Barrier Reef Foundation

1 This question asks for a named area. Only a country has been suggested. Whilst this is better than not putting any location, it is too vague. The answer could be improved by giving a specific location, for example, the Great Barrier Reef, which the rest of the answer is referring to.

2 This part of the answer needs to include place-specific detail by referring to the Great Barrier Reef Marine Park Authority, not 'they'. The reader does not know who 'they' is referring to.

3 A better response would give specific examples of zoning on the reef. For example, 'The Great Barrier Reef Marine Park is divided into different zones. Each zone has different levels of protection and activity restrictions. This aims to balance conservation goals with sustainable use. For example, some areas are designated as no-entry zones to protect sensitive habitats whilst others allow tourists to scuba dive, but only whilst being guided'.

4 This section is mostly descriptive and only briefly outlines what has been done. It reads like a list. A better approach would be to use developed statements that explains why this is a sustainable approach, such as, 'In zones that allow activities such as scuba diving, tourists are reminded not to damage the reef as coral grows slowly and is very fragile'.

5 When using acronyms, you need to list the full name first and outline the acronym in brackets. This way, the reader knows what you are writing about. It would have been better to write, '...suffered from crown-of-thorns starfish (COTS)'.

6 The answer is listing impacts here, not explaining how the reef has been managed sustainably. Although there is some place-specific detail, this is not answering the question. Remember to stay focused on the question that has been asked, not the question you think it is, or indeed want it to be!

Now, write an improved response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics. This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
explain how the processes of erosion, transportation, and deposition shape the coast			
describe the characteristics and explain the formation of coastal landforms			
identify the opportunities and hazards of living near coasts			
understand the causes and effects of coastal erosion and flooding			
assess hard and soft engineering strategies and techniques to overcome coastal flooding and erosion			
evaluate strategies and techniques used to protect the coast from the impact of tropical storms			
explain the importance of coral reefs and mangroves and what threatens them			
evaluate the strategies and techniques used to overcome threats to coral reefs and mangroves.			



> Topic 3

Changing ecosystems

LEARNING INTENTIONS

By the end of this topic you will be able to:

- give the main characteristics of the Antarctic and tropical rainforest ecosystems
- explain the biotic and abiotic factors influencing the characteristics of the Antarctic and tropical rainforest ecosystems
- outline the uses of and threats to the resources from the Antarctic and tropical rainforest ecosystems
- explain the impacts of the destruction of the Antarctic and tropical rainforest ecosystems
- describe and evaluate the sustainable management of the Antarctic and tropical rainforest ecosystems.

GETTING STARTED

Working in pairs, take a few minutes to complete the following:

- 1 Create two spider diagrams, one with tropical rainforests at the centre, and one with Antarctica at the centre.
- 2 Complete the spider diagrams, noting down everything you know about tropical rainforests and Antarctica. Think of the types of plants and animals, the climate, and how humans impact them.
- 3 Then take a few minutes to discuss the changes taking place in tropical rainforests and Antarctica.

Now in small groups:

- 4 Considering these two ecosystems, discuss what you think we could do to try and slow down the change that tropical rainforests and the Antarctic are currently experiencing.